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Unlocking the black box: Enhancing human-AI collaboration in high-stakes healthcare scenarios through explainable AI

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ABSTRACT

Despite the advanced predictive capabilities of artificial intelligence (AI) systems, their inherent opacity often leaves users confused about the rationale behind their outputs. We investigate the challenge of AI opacity, which undermines user trust and the effectiveness of clinical judgment in healthcare. We demonstrate how human experts make judgment in high-stakes scenarios where their judgment diverges from AI predictions, emphasizing the need for explainability to enhance clinical judgment and trust in AI systems. We used a scenario-based methodology, conducting 28 semi-structured interviews and observations with clinicians from Norway and Egypt. Our analysis revealed that, during the process of forming judgments, human experts engage in AI interrogation practices when faced with opaque AI systems. Obtaining explainability from AI systems leads to increased interrogation practices aimed at gaining a deeper understanding of AI predictions. With the introduction of explainable AI (XAI), experts demonstrate greater trust in the AI system, show a readiness to learn from AI, and may reconsider or update their initial judgments when they contradict AI predictions.

1. Introduction

Recent AI advancements have expanded the scope of algorithmically completed tasks, making it common for knowledge workers to use algorithmic support (Allen and Choudhury, 2022; Anthony, 2021) and potentially transforming their professional judgment capabilities (Choudhary et al., 2023; Townsend et al., 2024). However, the predictive capabilities of AI systems are often overshadowed by their opacity, leaving users puzzled about the logic behind their outputs. This 'black box' nature of AI significantly hampers user trust, error detection, and accountability, particularly in critical decision-making scenarios such as medical diagnoses (Kim et al., 2023; Meske et al., 2022; Meyer et al., 2024). As AI continues to challenge human expertise, the pursuit of XAI becomes crucial for bridging the understanding gap between humans and AI systems (Allen and Choudhury, 2022; Haque et al., 2023; Lebovitz et al., 2022). Despite this, the involvement of healthcare professionals in this rapidly evolving field remains insufficiently examined (Wang et al., 2024). Furthermore, AI research in healthcare suffers from fragmentation and a persistent divide between technological advancements and their practical implementation in clinical settings (Meyer et al., 2024).

Recent research examines how explainability influences humans' perceptions, attitudes, system utilization, trust, and information processing (Bauer et al., 2023; Erlei et al., 2020). Organizational researchers investigate how professional knowledge workers experience and manage the opacity of AI when making judgments (Glikson and Woolley, 2020; Van Den Broek et al., 2021; Waardenburg et al., 2022). Although several scholars argue for limiting the use of black-box AI in critical decision-making contexts (Amann et al., 2020; Burrell, 2016; Rudin, 2019; Teodorescu et al., 2021), we align with an alternative perspective that recognizes the inevitability of AI integration in high-stakes domains and seeks to optimize human-AI collaboration rather

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than restrict it. This position builds on the observation by Lebovitz, Lifshitz-Assaf, and Levina (2022, 140) that "explainable or interpretable AI may enable but not guarantee that professionals are able to integrate AI knowledge," highlighting the need for experts to reconcile divergent knowledge by aligning their own claims with AI claims through interrogation practices. While previous research offers important insights, it has not addressed how XAI updates human judgments in high-stakes scenarios (e.g., medical emergencies) where expert judgment diverges from AI. Additionally, the literature rarely examines the critical transition between opaque and explainable states and its impact on expert cognitive processes and decision confidence. This gap is particularly concerning in healthcare contexts, where effective human-AI collaboration could significantly improve outcomes. We address this gap by exploring how experts using AI systems deal with opacity and explainability, and what implications these factors have on their initial judgments. Therefore, we ask the following question: ***How do human experts react to the explainability of AI systems after experiencing AI opacity in critical judgment formation?*** We seek to understand whether XAI can facilitate collaboration between humans and AI systems in critical decision-making.

To address this research question, we conducted 28 scenario-based, semi-structured interviews and observations with clinicians from Norway and Egypt, whom we refer to interchangeably as "human experts" in our study. We examine these experts' interactions with an AI system developed using machine learning (ML). Our terminology follows the EU expert commission (European Commission, 2019), which defines ML as a subfield of AI. Specifically, we employ a Graph Convolutional Network (GCN) developed using deep learning techniques. The model predicts the risk of cerebral palsy (CP) by analyzing videos of infants and extracting spatial and temporal body movement features.

Our data analysis reveals that, during the process of forming judgments, these human experts enact AI interrogation practices when faced with opaque AI systems. When they understand the explainability of AI systems—specifically, how AI identifies potential disease risks and interprets results —they respond with varied reactions such as curiosity, asking questions and making reflective comments. This engagement leads to increased AI interrogation practices aimed at obtaining a deeper understanding of AI predictions. With the introduction of XAI, these experts demonstrate increased trust in the AI system and show a readiness to learn from it. Moreover, our analysis reveals that human experts may doubt their own competence when their judgments contradict AI predictions, prompting them to reconsider or update their initial judgments. Although the introduction of XAI encourages experts to recognize the potential of AI systems to identify nuances and offer standardized evaluations, we argue that overreliance on AI for such evaluations could lead to learning myopia within organizations, ultimately limiting both individual and organizational learning capacity.

We contribute to the current dialogue in organizational science regarding human-AI collaboration and opacity (Allen and Choudhury, 2022; Lebovitz et al., 2022; Waardenburg et al., 2022) by exploring influence of AI opacity and explainability on the formation of clinical judgments within the healthcare context. Our study advances this discourse by highlighting the relationship between AI opacity and epistemic uncertainty (Griffin and Grote, 2020; Oberkampf et al., 2004; Packard and Clark, 2020), demonstrating the various methods through which AI interrogation practices can occur, thereby substantiating the findings of Lebovitz et al. (2022). We extend this discussion by incorporating XAI, showing how it can alleviate uncertainties associated with opaque AI systems (Jiang et al., 2022; Tomsett et al., 2020). Additionally, drawing on our empirical findings, we propose a set of guidelines for designing human-AI interactions that support judgment formation and aim to maximize the benefits of human-AI collaboration.

2. Theoretical background

2.1. AI opacity and explainability in human-AI collaboration

AI systems that provide predictions for critical decisions, such as medical diagnoses, often function as black boxes, making it challenging to understand how the algorithm reaches a particular outcome. Current literature describes AI opacity as a lack of explanations for specific decisions, particularly for users with limited technical expertise (Glikson and Woolley, 2020). The characteristics of AI opacity can result in significant drawbacks, such as reduced user trust, compromised error safeguards, limited contestability, and diminished accountability (Bauer et al., 2023). The lack of explainability can lead to algorithm aversion (i. e., noncompliance with algorithmic advice), as individuals may not trust the black box system's ability to execute tasks accurately (Allen and Choudhury, 2022). Enhancing the AI system's transparency and human control enables individuals to understand and responsibly use the system (Dietvorst et al., 2018), whereas a lack of transparency can erode trust and lead to hesitation in adopting new AI services (Kim et al., 2023).

Transparency is a critical issue in ML, which involves analyzing large datasets and detecting correlations between data points to create predictive models, identify rare items, events, or observations in the data, or generate new data points. These predictions can surpass human capabilities and contribute to the expansion of human knowledge. However, a central challenge associated with advanced ML models— most prominently neural networks— is the difficulty of understanding how and which correlations between data points are used, making it challenging for humans to comprehend the reasoning behind ML model predictions (Haque et al., 2023; Waardenburg et al., 2022). This opacity, often referred to as the AI black box, has been shown to provoke frustration and disappointment among users, as they struggle to grasp the rationale behind AI results (Lebovitz et al., 2022).

Knowledge work collaboration involves integrating others' knowledge, which requires understanding and potentially revising one's own knowledge claims while critically evaluating others' input (Carlile, 2004; Levina, 2005). Similarly, human-AI collaboration entails the shareing of diverse knowledge, expanding both human and AI knowledge bases. Raisch and Krakowski (2021) describe this as a tight coupling, where machine outputs prompt humans to reconsider their views, and human input refines ML processes. This collaboration allows for varying levels of autonomy and interdependence between AI and humans, enabling the emergence of novel solutions (Sierhuis et al., 2003). In knowledge work, sharing is often difficult due to tacit knowledge and resistance to new ideas (Ambrosini and Bowman, 2001; Carlile, 2004). Sharing knowledge becomes especially challenging when the reasoning behind an argument is ambiguous, particularly across different fields. Likewise, when confronted with a contrasting opinion from an unexplainable AI, individuals may struggle to understand and accept new perspectives. The black-box nature of AI can pose a significant challenge when combined with contradicting cognitive claims of human expert (Lebovitz et al., 2022).

2.2. Explainable AI

The evolution of ML, accompanied by the increasing complexity of neural networks, has led to a growing focus on XAI. Its role is crucial in ensuring that AI systems provide sufficient information about their operations to end-users. Interest in XAI is particularly pronounced in safety-critical domains, including healthcare, where it is expected to foster trust, mitigate bias, and aid in AI development (Doshi-Velez and Kim, 2017; Lipton, 2018). XAI has shown potential in boosting clinicians' confidence in AI predictions, enhancing diagnostic capabilities, predicting disease prognosis, and supporting planning and resource allocation (Letham et al., 2015; Loh et al., 2022).

The importance of XAI lies in its potential to improve human-AI collaboration, facilitate knowledge transfer, build trust in AI systems, justify actions, enhance usability, and identify errors—especially in critical domains like medical diagnosis (Caruana et al., 2015; Kohl et al., 2019). Furthermore, explainability can mitigate the risks associated with misinterpretating ML model outputs due to a lack of comprehension, as well as reduce potential errors in AI decisions, which can have severe consequences (Rana et al., 2022).

Determining what constitutes an acceptable explanation for AI systems is challenging due to inconsistent terminology, as terms like interpretability, transparency, and understandability are often used interchangeably (Doshi-Velez and Kim, 2017; Laato et al., 2022). Practitioners use up to 20 synonyms for XAI (Brennen, 2020), highlighting its conceptual ambiguity. Arrieta et al. (2020, p. 85) conceptualize XAI as follows: "*Given an audience, an explainable artificial intelligence is one that produces details or reasons to make its functioning clear or easy to understand.*" suggesting that an explanation's clarity and comprehensibility depend entirely on the audience to whom it is presented.

XAI methods aim to make AI decision processes understandable to humans by revealing the "why" behind specific predictions, thereby addressing issues such as distrust, lack of accountability, and error safeguarding (Bauer et al., 2021; Bauer et al., 2023). These methods seek to balance high predictive performance with clarity and understandability and have been the focus of extensive research in recent years (Brennen, 2020; Laato et al., 2022; Meske et al., 2022). In the context of this study, we align with previous scholars (Arrieta et al., 2020; Bauer et al., 2023; Doshi-Velez and Kim, 2017) in defining XAI as a method that clarifies outputs and provides details or reasons to enable human understanding of an AI system's functioning. While several studies highlight the importance of XAI (Chen et al., 2022; Meske et al., 2022; Phillips et al., 2021; Wolf, 2019), key questions remain about how users responded to modern AI explanations—particularly since human behavior often introduces unforeseen consequences, even in well-designed information systems (Chatterjee et al., 2015; Willison and Warkentin, 2013). Moreover, research has insufficiently examined how explanations influence users' situational information processing, i.e., their use of context-specific information (Bauer et al., 2023). There is also limited understanding of human reactions to XAI in high-stakes decision-making, especially when expert judgment conflicts with AI predictions.

2.3. Judgment and epistemic uncertainty

Judgment is broadly recognized as an individual's evaluation, estimation, and inference regarding the relationships among objects, options, entities, or actions (Bitektine, 2011; Tsoukas et al., 2024). Judgment differs from decision-making, which refer to the process of selecting a course of action, judgment involves integrating multiple, probabilistic, and potentially conflicting cues to evaluate a situation (Goldstein and Hogarth, 1997; Tsoukas et al., 2024).

A critical factor influencing both judgment and decision-making is uncertainty, which arises from incomplete or missing information, conflicting evidence, and dynamically changing variables (Choudhary et al., 2023). Researchers often link the study of decision-making and judgment to uncertainty (e.g., Townsend et al., 2024; Tsoukas et al., 2024). Uncertainty can be classified into two types: (1) aleatoric uncertainty, also known as irreducible or inherent uncertainty, and (2) epistemic uncertainty, also referred to as reducible or state-of-knowledge uncertainty (Oberkampf et al., 2004). Decision-makers often face situations involving both mitigable (epistemic) and immitigable (aleatoric) types of uncertainty (Packard and Clark, 2020). In this study, we focus on epistemic uncertainty, which is caused by a lack of data or knowledge, and can be reduced by obtaining more information (Tomsett et al., 2020). This focus is motivated by the manageability of epistemic uncertainty and recent research highlighting the significance of users' epistemic uncertainty in understanding the impacts of XAI. For example, Jiang et al. (2022) demonstrated that providing a "prediction rationale," which helps users connectthe indicators of a medical issue with AI advice, is especially beneficial when epistemic uncertainty is high. Such explanations can compensate for users' lack of knowledge, helping them make judgments under uncertainty. This highlights the importance of understanding human reasoning in the design of XAI and the necessity of considering context to evaluate XAI's impacts. When users face high epistemic uncertainty, XAI designs that explain the "why" behind AI advice tend to be more beneficial than those that explain the "how" (Jiang et al., 2022).

Our study investigates how human experts react to XAI after experiencing AI opacity in critical judgment formation. To explore this, we have designed a scenario-based methodology in which human experts interact with an AI system designed to predict the risk of CP in infants. During these interactions, we introduce XAI generated explanations and observe and analyze their reactions and responses. A detailed description of the methodology is provided in the following section.

3. Method

3.1. Research setting

CP is the most common cause of childhood disability and causes functional limitations and co-occurring impairments due to injury to the developing brain (Korzeniewski et al., 2018; Novak et al., 2017). Diagnosis is typically made at 12–24 months of age, and more than half of infants who develop CP have neonatal risk factors, such as prematurity, perinatal asphyxia or neonatal ischemic stroke (Walstab et al., 2004). Follow-up of high-risk infants requires extensive specialized medical special services, impacting the child, the family, and the healthcare system. There is no single diagnostic test;instead, clinical diagnosis is based on a set of criteria and symptoms, often identified by multidisciplinary clinical teams who examine the child multiple times during the first years of life. Current early detection methods—based on observational movement analysis, such as the General Movement Assessment (GMA) technique, and neurodevelopmental clinical tests—are highly dependent on experienced and trained clinicians, limiting their widespread clinical use (Silva et al., 2021).

In this study, we focus on an AI system designed to assist clinicians in the early detection of CP before five months of age, thereby supporting clinical decision-making. The AI system identifies a high risk of CP by analyzing infants' spontaneous movements in 3- to 5-min-long video recordings captured between 12 and 18 weeks post-term age. Early detection of infants at high risk of CP may enable timely interventions during infant development, reassure parents of infants who do not have CP, and improve healthcare efficiency (Silva et al., 2021).

The AI system for early CP risk assessment employs spatial temporal graph convolutional networks (ST-GCN), a specialized form of deep neural network (DNN), that extracts new features from graph data, potentially enhancing performance in complex decision-making scenarios, including human movement detection (Groos et al., 2022). An ST-GCN ensemble with the appropriate architecture has been shown to identify relevant movement features in young infants that are indicative of CP. Despite the proficiency of such models in complex decision-making, they fall short in self-explanation, as they are not designed to explain the processes behind their decisions or the features they aggregate to users (Phillips et al., 2021). Therefore, developing XAI techniques is crucial to making neural network models reliable, transparent, and accountable in clinical settings.

The AI system used in this study employs XAI techniques to provide temporal and spatial explanations of an infant's spontaneous movements in the video recordings, helping clinicians understand which parts of the input data contributed to the model's prediction of CP risk . Temporal and spatial explanations refer to methods and techniques used to understand and interpret the behavior of ML models over time and in different locations, respectively. Incorporating spatial explanations is crucial for analyzing the spatial variability in infants' spontaneous movements.

The interviews were conducted using illustrative videos and static images to communicate the system's conceptual and possible functionalities (illustrated in Fig. 1). Clinicians were presented with XAI-generated explanations through the following elements: A. It was explained that the model uses nineteen body keypoints tracking the infant's movements. B. The predicted window CP risk value between 0 and 1 was displayed continuously along and synchronized with the video of the infant. C. The final CP risk for the entire video was presented, expressed as a value between 0 and 1. D. Segments of the video with low- and high-risk scores highlighted were presented in the form of a timeline beneath the video of the infant. E. Examples of segments with high predicted CP risk were shown on the timeline. F. An illustration of Class Activation Mapping (CAM) was provided, showing areas associated with high predicted CP risk (in red) and low predicted CP risk (in green), based on the model. G. Figure illustrating the model's uncertainty, derived from the outputs of 70 individual models, was also presented.

The utilized XAI method CAM (Zhou et al., 2016) produces a localization map indicating the importance of the model's input features—in our case, the infant's 19 body keypoints. This is achieved by applying Global Average Pooling to the feature maps extracted from the last convolutional layer, which are then fed directly into the activation function to obtain the final class scores. For any given class, a weighted sum of the feature maps is then multiplied by the weights of the classification layer to obtain the final CAM values, which are used to assess and visualize the importance of the model inputs. CAM has previously been studied as a method for generating explanations in CP prediction models (Groos et al., 2022; Pellano, Strümke, Groos, Adde and Ihlen, 2024a).

3.2. Scenario-based method

As we categorize research on XAI and human-AI collaboration as a nascent research phenomenon, we opted for a qualitative research design that interprets open data for meaning (Edmondson and Mcmanus, 2007). We employed a scenario-based methodology using hypothetical scenarios—an approach common in social psychology and human-computer interaction (HCI) research—to explore participants' perceptions and decisions (Lee, 2018; Park et al., 2021; Petrinovich et al., 1993).

Although prior research suggests that scenario-based responses typically mirror actual behavior (Woods et al., 2006), we chose a scenario grounded in a real-life infant video used in GMA training to more accurately reflect clinicians' experiences. This decision aligns with calls to complement scenario-based studies with those involving real-life experiences to enhance the validity and authenticity of our findings (Lee, 2018).

Recent HCI research, particularly in XAI, has proposed and adopted scenario-based AI design, leveraging potential use cases early in product development and user research (Andres et al., 2020; Wolf, 2019). Scenarios play a crucial role in research by bridging the cognitive aspects explored in traditional HCI methods with the organizational focus of information systems development (Carroll, 1996; Wolf, 2019).

3.3. Data collection

We used interviews and observations to examine how clinicians deal with and react to AI opacity and explainability within their environment and during interactions with AI technology. Clinicians with practical experience in the early neurodevelopmental assessment of infants at risk for CP were interviewed. Some, but not all, participants had knowledge of and training in the use of GMA in clinical practice and/or research. We employed purposeful sampling (Suri, 2011), selecting participants who could provide rich insights into the phenomenon under investigation. Initially, we planned to collect data exclusively from a university hospital in Norway, leveraging a co-author's professional connections to access a group of 12 clinicians. These participants were selected for their direct involvement in evaluating infants with medical risk factors for CP. However, we recognized a potential limitation: informants might exhibit increased trust in the AI system due to their familiarity with and respect for the co-author's expertise. To address this concern and improve the generalizability of our data, we recruited five additional clinicians from other hospitals in Norway. This step expanded the dataset and reduced the risk of partiality.

Given the limited number of clinicians specializing in CP diagnosis, we further broadened recruitment beyond Norway. The developed AI model for CP prediction is intended for use in all settings where high-risk

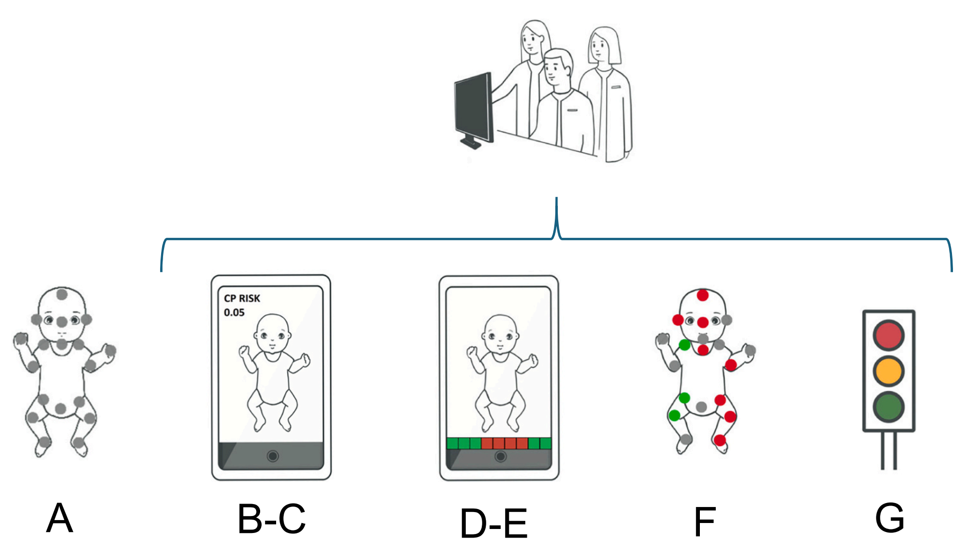

Fig. 1. Visualization of XAI elements presented in the interview.
Visualization of the different elements presented in the XAI part of the interview: A) Nineteen body keypoints tracking the infant's movements, B) The predicted window CP risk score displayed continuously along and synchronized with the video of the infant, C) The predicted final CP risk score for the entire video, D) Segments of the video with low- and high-risk scores highlighted in a timeline beneath the video of the infant, E) Examples of segments with high predicted CP risk on a timeline, F) An illustration of Class Activation Mapping (CAM) showing areas associated with high predicted CP risk in red and low predicted CP risk in green, G) The model's certainty level. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

infants are assessed, and the use of—and trust in—AI varies significantly between these settings. We selected a university hospital in Egypt as the second site, considering the stark contrasts in AI adoption between the two countries. Norway is a leader in AI integration, whereas Egypt is still in the early stages of implementing such technologies (Berntsen, 2024; Makhlysheva et al., 2022; OECD, 2024).This selection further expanded our dataset by incorporating diverse perspectives, allowing us to explore potential differences in clinician trust and perceptions of AI systems. For instance, we examined whether informants from Egypt, with limited exposure to AI in healthcare, might exhibit different trust levels compared to Norwegian informants, who work in a system where AI is more established. However, our data analysis did not identify any such differences.

In total, we conducted scenario-based interviews with 28 clinicians, comprising 17 from Norway (referred to as Nor) and 11 from Egypt (referred to as Egy), of whom 25 were female. The primary criterion for inclusion in our sample was experience in evaluating infants with medical risk factors for CP, representing potential end-users of the assessment tool within follow-up programs or outpatient clinics dedicated to high-risk cases. The sample included pediatricians specializing in neonatology and neurology, as well as physiotherapists and occupational therapists, encompassing a diverse range of professionals involved in the early detection and management of CP.

Our data collection also involved observing clinicians' reactions to the opacity and explainability of AI systems. The initial phase comprised 17 interviews conducted by two authors: one leading the interview and the other taking notes to document clinicians' reactions at various stages. In a subsequent round, a single researcher conducted 11 interviews, carefully recording key reactions to the transition from blackbox AI to XAI. This approach enabled us to capture nuanced insights and reactions from clinicians (e.g., smiling, signs of surprise or confusion) while interacting with AI technologies. Additionally, observation allowed us to document clinicians' interest in engaging with XAI and their questions about its inner workings. Furthermore, our observations allowed us to document clinicians' reactions to the AI's uncertainty metrics, which offered insight into the AI's confidence levels in its predictions.

3.4. Materials

We selected three videos of infants' spontaneous movements recorded for the assessment of CP risk, each lasting 3 min. The infant in Video A exhibited normal movement patterns, as observed by experts, while the infant in Video B showed clearly abnormal movement patterns. Video C presented a more challenging case, demonstrating both normal and abnormal movement activities. This approach aimed to increase the likelihood of observing contradictions between clinicians' observational judgments and AI predictions in at least one of the videos, allowing us to capture their reactions when their judgment aligned with or contradicted the AI predictions. The videos were authentic, previously assessed by human GMA experts, and CP status at 12 months corrected age or older was also available.

Before starting the main series of interviews, we conducted a pilot interview to assess our videos and interview guide, ensuring a unified understanding of the process. Based on the pilot interview and initial feedback from the clinician, we replaced two initial videos with three new ones of varying difficulty to better meet our research objectives.

3.5. Procedure (interviews and observations)

We conducted approximately 45-min semi-structured interviews and observations as part of a multi-phase data collection process. The interviews were carried out across three phases, capturing clinicians' reactions during and after each phase.

In phase one, we provided clinicians with an overview of the AI system and its specific application in our study. This step ensured that all clinicians received standardized information about AI in general, as well as the particularities of our AI system. In phase two, we presented clinicians with three videos without disclosing the results of the GMA or any clinical assessments. We asked them to identify any concerns that might suggest a preliminary clinical conclusion of “at risk” or “not at risk” for CP for each video.

In phase three, we presented clinicians with the black-box AI's overall predictions for the same videos, revealing only the final classifiications of each infant video as either “at risk” or “not at risk” for CP, without providing any explanation. We carefully observed the clinicians' reactions to the AI predictions. Following the observations, we conducted interviews, asking specific questions about their perceptions of the AI predictions—both when the AI predictions aligned with their own assessments and when they differed. We also asked whether the clinicians would readily accept the AI's predictions as presented or if they felt the need for additional information or explanations. Finally, we sought insights into their responses when the AI's predictions diverged from their own and how they viewed such situations.

In the final phase, we provided the clinicians with explanations of how the AI system works, including which parts of the infants' bodies the AI relied on for its predictions and detailed information about the training data used to develop the AI system. In cases where clinicians did not seek explanations, we asked whether they felt a need for additional information about the AI system, its functioning, or how it arrived at its predictions. Most often, the response was affirmative, highlighting the importance of understanding the AI's decision-making process. We then provided the requested explanations. Throughout this phase, we carefully observed clinicians' reactions to the explanations they received. The purpose of these observations was to assess their level of engagement and comprehension of the AI predictions when supported by XAI information. Additionally, we closely observed whether the availability of explanations enabled deeper discussions, prompted more questions, built trust, and encouraged integrative practices.

3.6. Opening the AI black box

The complete process of AI-based prediction of CP has been previously published (Groos et al., 2022) and is illustrated in Fig. 2. Initially, an infant movement tracker automatically identifies 19 body key points in each frame of the video, forming the structure of an infant skeleton model.

In this deep learning–powered method, movement tracking (step 1) involves using an image-based deep learning model to track infant movements from a video, creating a skeleton sequence of 5-s windows (step 2). Separate deep learning models then predict the risk of CP based on the body key-point movement data within 5-s windows. This is achieved by initially detecting movements of individual joints over a few time steps in the initial layers of a model, followed by the recognition of entire body movements across many time steps in the model's subsequent layers (step 3). The median prediction across 70 different models represents the CP risk (from 0 % to 100 %) associated with the 5-s window. The median CP risk value across all 5-s windows of a video is compiled to categorize an infant as either having CP or not (step 4), using a predefined decision boundary (dashed line at CP risk value of 0.35). This process also reveals the decision-making uncertainty through color coding based on the consensus among the 70 different models. The colors red (>75 % agreement) and orange (>50 % agreement) represent certain and uncertain predictions of being “at risk for CP,” respectively, while yellow (≤50 % agreement) and green (<25 % agreement) represent uncertain and certain predictions of being “not at risk for CP,” respectively (Groos et al., 2022).

3.7. Data analysis

In our analysis, we documented key points and main observations from each interview immediately after its completion. Following Braun

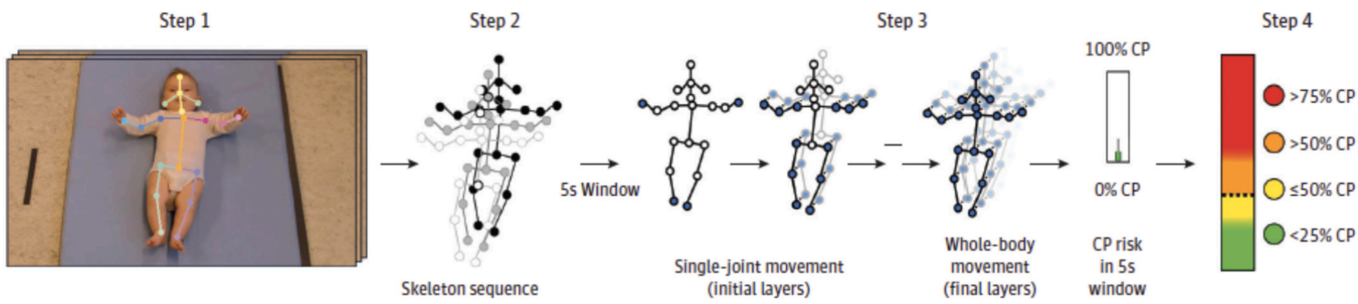

Fig. 2. Steps involved in deep learning–based method for cerebral palsy prediction. Note. The image was reprinted from Groos et al. (2022).

and Clarke's (2006) method for thematic analysis, and using NVivo software and Excel, we began by familiarizing ourselves with the data, generating initial codes, and searching for themes across the interviews.

Our initial objective was to understand the data and gather diverse viewpoints to ensure objectivity and minimize potential bias. After to each interview, we conducted initial coding, during which detailed notes were taken, and main insights summarized. The first three authors independently analyzed subsets of the interviews, each employing theme-based coding to identify emergent themes. Following this initial analysis, these authors met to review and refine the initial codes. They then worked separately once more to further examine the data, before reconvening for a comprehensive discussion. This multi-stage approach was important for capturing the breadth of the data and minimizing bias, in line with Eisenhardt's (1989) caution against drawing premature conclusions. Through these discussions, we reached a consensus and ensured a thorough understanding of the data. Subsequently, initial insights and key themes were recorded in an Excel sheet, providing flexibility to associate each reaction with its corresponding phase in the interview.

At this stage, our codes were descriptive, intended to accurately capture how clinicians described and felt about their experiences with AI as a black box (Jiang, 2021; Locke, 2001). We coded for their descriptions of and uncertainties regarding AI opacity and the lack of medical history. Clinicians expressed discomfort in forming judgments without access to the medical history of the infants or having direct interaction with them. This led to the identification of common themes, such as seeking direct interaction with infants and seeking access to medical history. As we progressed through the manuscripts, we began to notice other recurring themes related to reactions to the black box—such as searching for explanations for the AI's opacity, doubting initial judgments, and seeking the opinions of human experts.

We then thoroughly examined the initial themes identified in the previous steps and recoded the interviews using our inductive codebook, drawing upon Braun and Clarke's (2006) strategy for defining and refining themes. This involved identifying overarching patterns and discussing code relationships, grounded in our initial findings and the relevant literature on AI and opacity and explainability in healthcare (Jiang et al., 2022; Lebovitz et al., 2022). Following the descriptive coding approach (Jiang, 2021; Locke, 2001), we identified common themes in the manuscripts, such as experiencing AI opacity in use, enacting epistemic uncertainty, and reactingto AI explainability. For example, we wanted to understand what made clinicians uncomfortable making judgments without access to medical history or direct interaction with the infants, as well as the reasons why they updated their initial judgments, particularly when no AI explainability was provided. Additionally, we sought to understand how clinicians perceive AI explainability, why it prompts trust, and what implications it has for their final judgments. Consequently, we coded for recognizing AI's potential to reduce uncertainty, willingness to learn from AI, increased trust in AI, and the reconsideration or updating of initial judgments. Finally, the consensus coding phase marked the conclusion of our thematic analysis, during which discrepancies were resolved through discussion.

4. Findings

Our study exploreshow experts react to the explainability of AI tools after experiencing AI opacity during critical judgment formation. The findings are organized into two themes: (1) *Experiencing AI opacity*, which examines the practices experts use to interrogate AI predictions in the absence of explainability, and (2) *Experiencing Explainable AI*, which highlights how clinicians react to and incorporate insights from XAI into their judgments.

4.1. Experiencing AI opacity

In general, experts are willing to invest resource to obtain additional evidence that helps reduce their epistemic uncertainty, i.e. "ignorance of unknown but knowable information" (Lebovitz et al., 2022, 140; Packard and Clark, 2020) when making critical judgments. However, when clinicians encounter AI predictions, which can be considered additional evidence, their epistemic uncertainty often increases rather than decreases. This occurs because clinicians can not access the reasoning behind the AI predictions— they are experiencing AI-in-use opacity. In responseto this growing epistemic uncertainty, clinician chose to either enact AI interrogation practices or re-examine their own judgments.

4.1.1. Enacting AI interrogation practices

Our data analysis uncovered a range of reactions showing that experts enact AI interrogation practices—defined as "practices that human experts enact to relate their own knowledge claims to the AI knowledge claims" (Lebovitz et al., 2022, 133), when encountering AI opacity. These interrogation practices were observed in several distinct approaches:

4.1.1.1. Verifying AI claim with external sources. When presented with AI system outputs that contradicted their clinical judgment, clinicians expressed concern and preferred to double-check AI claims using either medical history or input from other experts. While they hesitated to fully agree with the AI, they did not dismiss its results outright. Instead, they sought medical history to better understand the results.As one clinician noted:"*In my work, the [video] isn't the only thing. I know something about the history of the baby. I need to put it into context*" (Nor17). Other clinicians also wanted to know the child's history and parental observations: "*I would obviously go through the history of the child. Were there any complications in the pregnancy, during birth? What have they observed in the first three months of life? What is the movement like at home? Have they noticed that one side of the body is used less than the other*" (Nor10). More details from parents were needed to make any judgment: *"I need a history from the parents. I need to examine the patient, [then], according to the history and the examination I build my concern or not […] I don't think [I can] depend only on a video, without history, without concerns from the parent, without anything without examination to build my [judgment]"* (Egy11).

Another way to double-check AI claims is to validate them with other experts—a common practice in medicine when facing epistemic uncertainty (here due to AI opacity).As one clinician noted: "*I think that is normal to do in medicine. I would consult some other clinician with more experience. As I don't get a more detailed answer from the AI [and I cannot] understand the system, I would probably ask a colleague, another doctor with more experience in this field, and I would, as we usually do, consult more experienced people"* (Nor1). Several clinicians expressed similar approaches. For instance, one said: *"If I was supposed to give the diagnosis or to tell the parents if my opinion were different, I would first see it once more myself and then I would check with an expert I think"* (Nor6). Another added: *"When I am not sure about the baby's movement or I see this movement as suspicious for me, I will refer this baby to a neurologist [...] I would refer to a more specialized doctor to a neurologist"* (Egy9). This practice of checking AI claims that contradict their judgment — either by consulting medical history or other experts—demonstrates that clinicians neither fully trust AI claims nor are willing to dismiss them outright. .

4.1.1.2. Seeking explanation for AI predictions. Another reaction to increasing epistemic uncertainty when dealing with AI opacity is the desire to understand why AI makes certain predictions and to request explainability: "*Why does AI think that there is a risk? And what did he [the baby]do wrong[...]I will always need to know why. I mean, if the AI is going to say, I disagree or agree [with my judgment], then I will need to know why [...] if you say, no, I'm concerned about this baby, then you would need to give me a reason"* (Egy6). Other clinicians expressed similar concerns about AI predictions. One states*"Basically, you cannot totally accept or agree with something that you don't fully understand the mechanism and the underlying process for deciding this baby to be affected or not"* (Egy2),while another added, *"I just don't want AI to miss a pathology that I have an intuition about. Like one limb that I feel like that one doesn't seem to move yet,"* (Nor8). She further stated, "*When I have doubts [...] I would require explainability."* Egy10 explained why XAI is important to them: *"[Explainability] is really important. I build up[my assessment], that he is at risk, because his right hand wasn't moving. So, well, his frequency of movement wasn't so good for his age. So, if I had a disagreement between me and AI, I would like him to convince me why he sees the baby as at risk, and I would like to learn. So, if his opinion is right, I would like to learn"*. Egy1 also emphasized the important of explainability: *"[XAI] in our job in medicine is very, very important. We're not talking about magic."* Others explained how XAI can affect their judgment: *"If I can understand the equation of what's going on behind, I would accept it"* (Egy2). *"It would increase my trust. Because then I could see like the disagreements. It would be important for a clinician to see what the program is thinking and in which way it does its evaluations" (Nor4).*

4.1.1.3. Reflectively agreeing with AI claim. Other clinicians took a different approach by reflecting on their own judgment. For example, some chose to closely monitor the case: *"That's a case where it would make me more uncertain of what the result would be for this child, so I would keep a closer eye on it to see how the child actually develops"* (Nor2). Some clinicians expressed a lack of confidence when comparing their own judgment with AI predictions and tended to believe that the AI's assessment might be more accurate: *"Maybe I'm naïve [...] it's about how much clinical experience you have in the field. So as a less experienced clinician in this general movement assessment, I think I would be happy to have a system who helped me... I would tend to trust the AI more than my own judgment even though it was contradictory" (Nor1).*

They believed the AI had better vision than they did. Even if they didn't fully trust the AI's predictions, they often doubted their own judgment due to a lack of experience or training: *"I wouldn't trust the AI completely in the second one. Yeah. Mm-hmm. No, probably not. But I can be wrong. I have a feeling that AI is better than me"* (Nor8), *"Maybe I need more training"* (Egy5); *"Maybe [...] AI can see better than I can in that case"* (Nor15); *"I probably would have doubted if what I saw was correct. I would have been a little like, am I wrong? Or maybe not trusting my own knowledge"* (Nor9). Some clinicians were unsure of their judgment based on watching videos alone and felt that the AI was reliable enough to provide them with the information they needed: *"[I am] less concerned [after the contradiction], maybe because I think that the AI is good enough to identify, for example, CP. And I would have checked if the baby was more active with stimulation. But it's difficult [with only watching video]. I'm not quite sure because I know that in three minutes, I would never be sure of my own conclusion" (Nor15).*

4.1.1.4. Synthesizing AI and their own claims to form a new claim. Although puzzled by discrepancies between AI predictions *and* their initial judgments, some clinicians sought a middle ground to reconcile with AI predictions. For example, they justified their initial claims and aligned them with AI claims: *"When I saw the video, it was not like that I thought this is a very unhealthy baby, because when you asked me, you said at risk or no risk, and I said at risk but low risk. So, it's not like I said it's obvious that this baby would have CP. When it [AI claim] contradicted me, it doesn't feel like I have any [contradiction], I would think like, yeah, maybe it is right then that it's no risk"* (Nor3). Another clinician felt relieved when AI provided more information amid uncertainty: *"I'm very happy because I feel like in the second video, I was not sure if I'm concerned or not concerned. I feel like it's normal, but I take the safest side so if someone told me it's normal that's fine"* (Egy4). Some experts also chose to update their own judgments based on AI predictions. For instance, Nor13 said they would use AI predictions to decide their next steps: "*I think that's just interesting. And if I know from research that the AI has been quite good in the prediction of CP, then I would have followed up the patients more. If the AI says it's high risk, I would definitely let them [the patients] come back to my clinic."*

4.1.2. Re-examining their own judgment

Some clinicians chose to suspend their judgment or initiate a new evaluation to ensure they were doing their due diligence. For example, Nor2 confirmed they would suspend their judgment when encountering a conflicting AI prediction: "*I would consider that I may be wrong, the child may develop differently than I think, but it may also be that the AI is wrong, and I have to watch this child closely to see what happens. So, I would suspend judgment."* Meanwhile, Nor4 preferred to conduct a new evaluation when there was a disagreement between the AI and their own assessment : *"If it was my patient, I would like to do a new evaluation, just to be sure. So, in every case, where there is a disagreement, I would do a new evaluation."* Other clinicians wanted to rewatch the video to understand why the AI made a particular prediction: "*First I would rethink, and then I would maybe watch it once more. And then try with my knowledge about these movements, maybe try to understand why the computers says it's different from me"* (Nor6); *"Can I see your second [video]?"* (Egy10), *"Can I just see it again?"* (Egy6).

4.2. Experiencing explainable AI: Making sense of AI logics and updating critical judgment

Our data analysis reveals that the introduction of XAI insights triggers diverse reactions among clinicians and has implications for their final judgments.

4.2.1. Reaction towards explainable AI

4.2.1.1. Willing to learn from AI system. Following the provision of XAI insights, clinicians have demonstrated an inclination and readiness to learn from AI to enhance their clinical judgment: *"It's a good thing, or it could be used as a way to learn"* (Nor8). This is especially relevant given the scarcity of experienced clinicians specializing in CP and the infrequency of CP cases. Nor8 further elaborated:*"In an ideal setting, you'd have a really experienced doctor beside you, and you'd have lots of CP kids* *and get to learn that. But in reality, that doesn't happen—you have to teach yourself. And reading in books is not—it's way better to see it on video. So, I think it could be a part of a very good tool to learn about CP."* This response highlights a readiness to adopt the AI system as a supplement to conventional educational resources, positing it as "*a learning opportunity"* (Nor17). The participant further explained that it *"is best if you are using the AI also to sort of simultaneously train us a little bit to develop the human eye as well."* This embrace of the AI system as a pedagogical instrument was further articulated by Nor1, who reflect on the potential reliance on XAI to enhance judgment and enable learning, proposing a scenario: *"If I was a pediatrician on an island [...], and I had to do this CP prediction, which is maybe not so probable, because it may be done in a more centralized place—but how I would use it then? I would look at the video, and I would look at the colors, and the time. I would try to learn from it."*

Moreover, clinicians showed a readiness to learn from AI following the introduction of XAI: *"When I know a bit more about how these machines interpret the movements from these marks and see what happens on the timeline, then I could concentrate and see if, yeah, maybe in that period of the video, I agree, if it's less movements or whatever. So, I think that would be okay to sort of learn how to interpret videos like this"* (Nor14). This stance shows a willingness to validate AI predictions through interrogation practices, where clinicians invest time and analytical skills to align their expertise with AI insights. When a clinician's judgment differs from AI predictions, XAI can bridge the gap by allowing clinicians to reconcile their understanding with that of the AI and enhance their knowledge through such interrogation practices: "*If I had a disagreement between me and AI, you would like him to convince me why he sees that the baby is at risk, and I would like to learn"* (Egy10).

4.2.1.2. Recognizing AI system's potential to minimize uncertainty. Our data analysis shows that, following the introduction of XAI, clinicians began to view AI as a tool to help address uncertainties—espcially when they were unsure about their own judgments. In such cases, AI serves to challenge or confirm their judgments. Egy1 recognized AI's role in resolving ambiguity within initial judgments, stating, *"If the AI is sure this baby is at risk, while I am not sure, I should put this in my considerations."* The introduction of XAI can enhance clinicians' trust in AI predictions and reduce reliance on uncertain personal judgments: *"I definitely think I could be influenced by AI. If I was uncertain and you gave me like totally green [indication of no risk] on all extremities, then I would probably say, OK, fine, AI is better than me" (Nor8).*

The view of AI as a superior consultative agent that helps reduce uncertainty in initial judgments was also reflected in Nor3's statement: "*It would be in cases where I felt uncertain , because sometimes it could be that you examine a child, and you're not completely sure what this means… Then at least then it would help me a lot if it said the other way; then I would trust that more than my own examination."* Similarly, Egy5 emphasized their reliance on AI to confirm their clinical judgment in ambiguous situations, stating: *"We need to evaluate the baby first, then confirm with AI results.*" She elaborated on AI's role as *"an aiding tool for final confirmation of the diagnosis… to pick up the baby at high risk If we are confused, maybe aid until we take a consultation with the pediatric neurologist."*

4.2.1.3. Increasing trust in AI system. Upon the introduction of XAI, clinicians showed an increase in their trust towards the AI system. This shift was particularly evident when the inner mechanics of the AI were unveiled to them. Nor6 expressed greater confidence in the AI after being exposed to its operational logic: "*Absolutely, when you showed me how the computer is working, I think that's very smart… I trust it, I think it sounds very smart, or correct, the way it does it […] I feel in a way that I can trust it."* Nor1 also noted that XAI fostered more trust in the AI system, stating "*I would trust that much more, because it makes me able to understand a little more what the AI does."* Nor10's reaction to XAI insights was articulated as follows: *"It's useful to know more about how it works because it doesn't seem so random. And of course, knowing that my own knowledge is not infinite, I wouldn't have bet my life on my own predictions in these babies (smiling) [...] It makes me trust it more when I see how it has evaluated these movements"* (Nor10). Similarly, Nor14 stated, *"I have no self-confidence in cases like this. I know too little about how to interpret these movements myself. So, for now, I would trust the AI because I don't trust myself (smiling)."* These quotes illustrate experts' self-awareness and humility, acknowledging their limitations as humans while recognizing the capabilities of AI.

The introduction of XAI and the explicit articulation of AI's certainty levels in its predictions are essential for promoting trust among clinicians. For example, Nor2 highlighted the value of transparency in building trust, stating, *"Transparency and openness about what isn't certain, what we don't know, I think is probably very important for the level of trustworthiness for me."* In the same vein, Nor3 explained the impact of disclosing the certainty of AI predictions on their confidence in the technology, stating: *"If the uncertainty is not mentioned, then that would create more uncertainty. Because you know that it can't be that easy. But when it is mentioned, the uncertainty, that would make me trust it more."*

The reaction of Egy11 provided an illustrative example of the transition from initial skepticism to trust upon exposure to XAI: "*At first, I think I don't have to hear it from him [AI]. And I will only depend on my mind or opinion. But when I see how it works and how it goes by like five seconds and the 19 points […], I feel better, and I feel that I am happy to hear from it. It gives me more trust.* They expressed their happiness as we observed their positive reaction and growing trust in the AI system. Similarly, Egy4 conveyed greater trust in AI after understanding its underlying mechanism: *"After you explained to me how you trained this machine based on these points, and that he divides every video into five seconds. That's why his decisions are based on numbers. So, I think I still feel like I am happy with this, and I feel like I trust it much more after your explanation."* However, despite the growing trust in XAI, Egy3 tempered their confidence with the need for further empirical validation: *"It is giving me more trust, but still not 100%.... until it's approved by another studies that we can compare using AI by a database, I want to trust the database that was given to the AI."*

4.2.1.4. Enacting more AI interrogation practices. Our data analysis indicated that the introduction of XAI encourages clinicians to engage more deeply with AI. Clinicians showed engagement by expressing curiosity, asking additional questions, and providing reflective comments when they received XAI information. They specifically showed increased interest in re-examining infant videos paired with XAI insights, underlining their pursuit of a deeper understanding through the lens of XAI: *"AI estimates the overall risk…. Can I go back? [in the explanatory video] It considers the risk of CP and the overall risk, which is at what cut-off point… It makes sense. Yes. But what if he judges one point at one frame as not moving enough,and the next frame, the joint on the point is moving adequately? How is it correct…?Each model assesses what?"* (Egy2). In the same vein, Egy11 inquired about the circumstances under which the video recordings of the infants were made: *"When they ask the parents to get a video for the baby, at what times did they give the videos? I mean, at what time of the day...I'm asking because if the baby is ill, he will not act like he is normal. If the baby is hungry, if the baby has colic, he will not act like normal…. I feel more satisfied now that he depends on these points."* Nor3 asked, *"Does the AI have information about the things in the history that makes risk of CP? […] Or is the AI not fed with, like, this is a baby born premature?"* These inquiries and interrogations reflect clinicians' high level of interest and engagement with XAI— not only to reduce their uncertainty but also to align their knowledge claims with those generated by the AI.

Nor8 explored the specificities of the AI's operational mechanisms, asking, "*Does it take into account if the kid is completely stiff in the upper extremities but very active in the lower?"* She praised XAI for its clarity in identifying the affected body sides, stating, *"that's excellent to identify what type of CP and what side of the body. That's what I was hoping. So, you know that in the black box."* Her subsequent request, *"I would like to see* *that for the second child, how it evaluated the second child,"* reveals an interest in further exploring the AI's analytical process further, especially after gaining insights into its operational mechanics.

After observing red body points in the XAI videos, indicating the factors contributing to the AI's prediction of CP risk, Nor4 requested a second viewing of the infant's video. This clinician pointed out the value of revisiting specific segments highlighted by XAI, saying, *"It would be useful to go to these red parts and see the video again and again to see. So, I think it's quite interesting to have this bar."* This comment emphasizes the usefulness of XAI in enhancing clinicians' ability to assess CP early.

Furthermore, some clinicians probed deeper into their investigative approach towards AI by seeking detailed information about its training data. For example, Egy5 inquired about the scope of the training data, asking *"Was it [training data] national or international?"* Similarly, Egy7 asked *"How many infants were included in the study...that's important."* Egy6 questioned the validation process for the training data, stating, *"To believe that AI is actually predicting right, what I would expect is that these patients that you have told that they are at risk that you have followed with them and then actually they become cerebral palsy later on"* This participant also expressed concerns about the dataset's focus, pointing out, *"The other thing that I will be concerned about is that your data is based solely on the motor movement of children, right, which is not the only thing that I'm going to look at as an infant when I'm examining him. I need to look at the whole infant."*

4.2.1.5. Cautioning against sole reliance on AI system for decisions. While the introduction of XAI appears to boost clinicians' confidence in AI systems, it has also encouraged a cautious approach. Clinicians emphasize viewing AI as a complementary tool that enhances—rather than replaces—traditional clinical assessments. This perspective reflects an understanding of the capabilities of AI while recognizing the irreplaceable value of human judgment and thorough clinical evaluation: *"I would see it as a decision support…I would use it in addition to my own experience, to the history, to everything, the other values from the infants"* (Nor6). Similarly, Nor2 stated, *"I would never completely trust AI judgement and disregard my own completely, but I take everything into consideration…. We never make a judgement completely based on just one thing. We always sit down and consider the totality of it."* Nor10 also referred to the inherent limitations of predictive and diagnostic tools, including AI systems: *"It will always be only a tool, one of many tools that we have…We always check, we always measure…so far, most of our tools are flawed, and we cannot 100% trust them."* This clinician expressed cautious optimism— recognizing the benefits of XAI while emphasizing the need for diverse assessment methods: *"After providing this information, I feel more confident about AI as it depends on 19 areas for evaluation… but I still insist on another tool for assessment besides the AI"* (Egy8). This perspective was supported by Egy10, who acknowledged the strengths of the AI system while highlighting the importance of human decision-making: *"He gave me convincing evidence. But still, as I said, being an artificial system, he can't replace me. So, I am convinced by the evidence he gave, but still I have my own decision to make."* Egy11 summarized the cautious stance adopted by clinicians, stating, *"This is a good explanation… how it works is very good, but not enough to depend mainly on it."*

4.2.2. Refining judgment with XAI insights

4.2.2.1. Reconsidering initial judgment. XAI encouraged clinicians to rethink their initial judgments, driven by the heightened trust in AI that XAI fostered. This trust, in turn, was often accompanied by self-doubt, promoting clinicians to reconsider their initial assessments of the infants. Egy9 exemplified this shift, showing a willingness to align AI prediction with their expertise, stating: *"Now I will take the diagnosis of the AI into consideration, and I will rethink about it, and I may refer the baby for further intervention or investigation."* AI prediction also motivated clinicians to re-examine infants rather than being the sole decision maker: *"If I hear from him [AI] that he has concern and I do not have concern, I will examine the baby one more time. And If I have concern and he does not have concern, I will still be examining the baby again. If I see that I was wrong, this is good. So, it's good to have another opinion"* (Egy11). The realization that AI can challenge and complement the clinician's expertise is confirmed by Egy7: *"If it is a result against my opinion, I will take a second look at the baby—do more examination, [gather] more details about the history, more investigations—to decide whether this is truly at risk or not… Yes, I want to check the baby again."* Nor4 similarly emphasized this point :*"I would just do a new evaluation [for the contradictory video]."*

Incorporating AI into clinical workflows does not obscure the value of human judgment but rather adds an additional layer of analysis. Clinicians such as Nor12 and Nor8 viewed AI as a digital colleague, offering perspectives that could confirm or challenge their own. Nor12 emphasized the importance of reevaluating uncertain situations, stating:*"That would be very good. I think if I would use this, I would like this grey zone area. That makes it important to do it again or to reassess."* Nor8 valued the contribution of AI in enhancing clinical judgment, particularly in cases where personal confidence was low or when a case *"somehow came in this uncertainty region."* In such scenarios, Nor8 regarded AI predictions as *"nice with a confirmation from an AI."* She also stressed the need for further clinical evaluation in cases marked by doubt: *But the unsure cases, I don't trust myself. So that's why I also bring the kid in again."*

4.2.2.2. Updating initial judgment. The introduction of XAI has prompted clinicians to reassess and update their initial judgments. For example, Nor3 reflected on this influence, stating: *"It is [the reason for changing their judgment] what we are talking about—that I know that the algorithm put it at low risk and that I saw this score system [part of XAI], which made me think about what was with that baby that made me think that it had a risk."* This self-reflection on the clinician judgment process highlights the role of XAI in encouraging clinicians to reevaluate their risk assessment. For example, Egy1 explained how XAI influenced their clinical assessments, noting that their own assessments may overlook some of the nuances that the AI system detects: *"I may be not accurate; I may be not concentrating on some points that the AI was seeing. So, I will consider his opinion and then make it a point to start."* This recognition of potential human oversight emphasizes AI's role as a collaborator—complementing human expertise and contributing to more accurate judgments. Moreover, the insights provided by XAI resonate particularly with clinicians who acknowledge their own limitations in specific training: *"Because I am not trained, I do not consider myself as someone who's trained. So, I will not argue with something that is trained even if it's a machine. If I am trained as GMA, and there's a difference between me and the AI, I think it will be a different scenario at this point. But I'm not sure what my degree of confidence in myself will be at this point. But because now I'm not trained, and if I have only the AI in front of me, and don't have any other opinion, so I will rely on the AI not because I am trusting it 100% but because I'm not trusting my training"* (Egy5).

4.2.2.3. Reflecting on potential use of the AI system. With the introduction of XAI, clinicians recognized the AI system's potential to: a) identify nuances that might elude human detection, b) remember infants' movements more reliably than humans, and c) offer standardized evaluations across various body points simultaneously.

AI's ability to detect nuances that may escape human observation has emerged as a key advantage: *"It has a pretty good pickup rate, and it does see patterns or see things that I, as somebody who is fairly trained in pediatric neurology—but not in this method specifically—it could add something that I don't see"* (Nor10). Nor4 suggested that the usefulness of AI goes beyond assisting trained professionals, noting its potential to support less experienced clinicians by identifying abnormalities that might be missed by an untrained eye: *"Of course, if you are untrained or not experienced, I think it will be useful. And probably it will pick up some cases that maybe* *[someone]untrained would say, I think this baby is normal and maybe it's not.*" Further elaborating on AI's detailed detection capabilities, Nor4 shared a personal observation of the AI system pinpointing precise details: "*The program can pick up something, these small details that we cannot see. I think it could be—it's more specific than a human…I look at a child, but I didn't see—like here, it picked up something with the left arm, and I didn't see it so clearly. It would be helpful for me, because then I could see like this skeleton, and I could go back to the video, then I could see like more specifically to this part of the body.*"

Furthermore, Nor8 highlighted that XAI can recall specific behaviors and movements throughout a video —something that is challenging for humans: "*That's [XAI]very illustrative. Because when you are a human, it's really hard to remember that, 'Oh, the kid was acting normal in the beginning of the video. I just remember the one thing that I was. So, I'm not able to, as a human, process, you know, at least I'm not good enough.*" This recognition of human limitations in processing and remembering nuances of movement over time was further exemplified by their admission of difficulty in recalling specific segments of the evaluation: "*This is very demonstrative for me because it is sort of an average. And it's hard to remember how many five of those five second things were red or green in my human brain.*" (Nor8). Additionally, she valued AI's potential to enhance confidence and identify overlooked details, particularly in the presence of external distractions: "*I can admit, there are limitations when I'm distracted by the parents and other things. So, it would really add confidence and let me see things that I haven't seen.*" These reflections reveal a recognition of AI's capability to complement human memory and attention, offering a more detailed and thorough analysis of infant movements.

The ability to provide standardized evaluations and assess multiple body points simultaneously offers a unique advantage over the GMA method, which requires extensive training and can vary between practitioners. Egy3 noted that GMA's subjective nature can lead to different conclusions from different practitioners assessing the same patient. In contrast, an AI system could enhance accuracy and minimize such variations: "*GMA is an individualized method. I can assess the patient, and another one can assess the patient by the same method and give me different results. I think it will be more accurate and at this point. It will minimize that variation.*" The participant also highlighted AI's efficiency,emphasizing its ability to examine multiple body points simultaneously, thus surpassing traditional assessments in speed: "*It's faster to assess different points at the same time and would be faster than individual assessment. As I may examine the hand first, then the foot, then the head, but the AI may examine them all at the same time with a variation of five seconds between every movement, I think it is faster than human examination.*"

5. Discussion

5.1. Summary of findings

Our study investigates how experts respond to the explainability of AI systems after encountering AI opacity during critical judgment formation. We observed that experts first constructed a provisional diagnosis when assessing the babies' conditions in three videos scenarios. After receiving AI predictions for these three videos, they experienced AI-in-use opacity, as they did not understand the rationale behind the AI's outputs. This led to increased epistemic uncertainty, particularly when their own judgments contradicted the AI predictions. Fig. 3 provides an overview of this process and the different categories of experts' reactions. We categorized experts' reactions to AI opacity into two groups: (i) re-examining own judgment and (ii) enacting AI interrogation practices, such as verifying AI claims with external sources; seeking explanation for AI predictions; reflectively agreeing with AI claims; and synthesizing AI and their own claims to form a new claim. After receiving XAI, experts began to establish a connection with AI—for example,willing to learn from AI system, recognizing of AI tool's potential to minimize uncertainty, , increasing trust in AI system, enacting more AI interrogation practices, and cautioning against sole reliance on AI system for decisions. Following the provision of XAI, experts decisions fell into three categories: reconsidering initial judgment, updating initial judgment, and reflecting on the potential use of the AI tool.

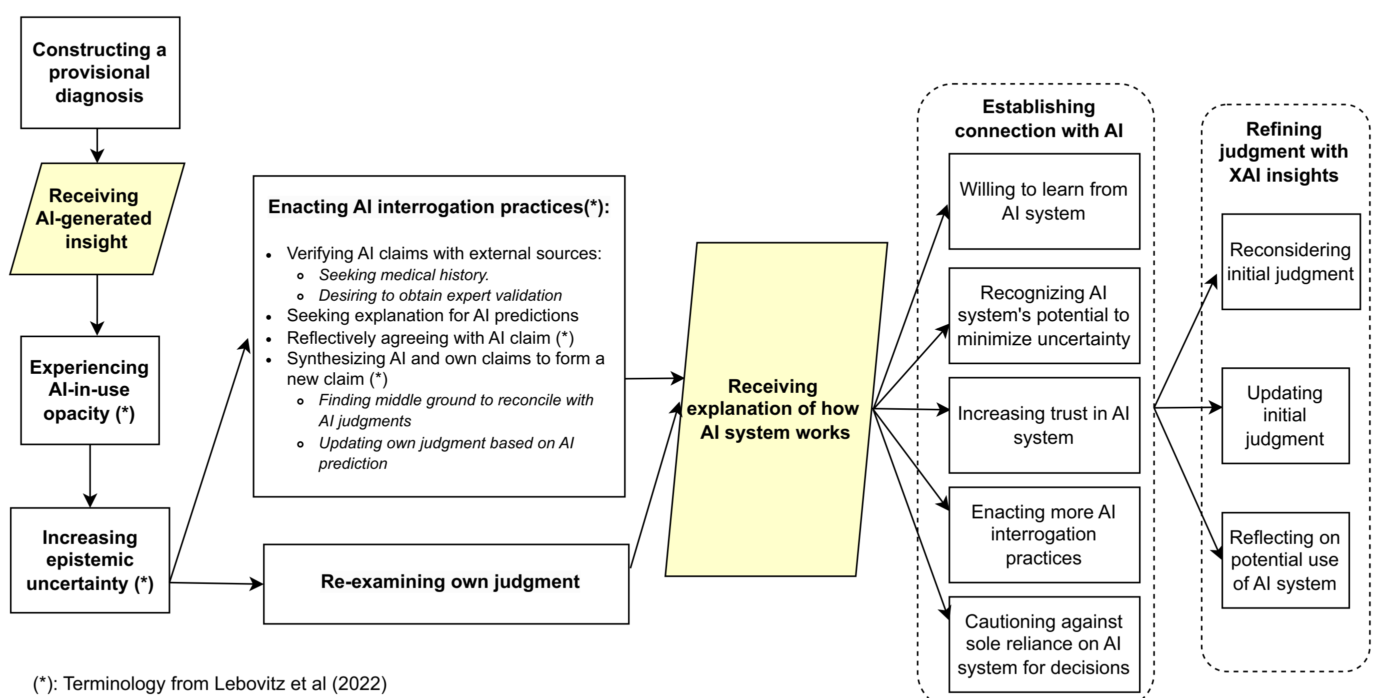

Fig. 3. Reactions of clinicians to explainable AI after experience AI opacity.

5.2. Theoretical implications

Our study confirms the speculation that introducing XAI motivates experts to engage more thoroughly with AI, as evidenced by increased AI interrogation practices and greater interest in XAI. This enhanced engagement manifests as a notable increase in curiosity, with experts asking more questions and offering reflective comments when interacting with XAI outputs. Specifically, they express interest in revisiting infant videos with XAI insights, indicating a desire for deeper understanding through XAI. These inquiries reflect experts' strong interest in using XAI to harmonize their professional insights with AI knowledge claims. This aligns with previous research suggesting that professionals benefit more from engaging with AI through interrogation practices rather than merely accepting or rejecting unexplained AI claims. There is a growing push for AI systems design that facilitates the integration of AI knowledge into professional practices, and it has been speculated that explainable AI may support this integration (Lebovitz et al., 2022). Our findings confirm that the provision of XAI does not deter experts from engaging in AI interrogation practices; rather, it encourages them. Moreover, our analysis uncovers a tendency among clinicians to rely on AI as a strategic response to the inherent uncertainties in diagnosing CP. AI is used to validate and question their prognostic decisions, leading to the perception of AI as a superior advisory entity that can minimize uncertainty in preliminary judgments.

We also found that expert motivation to learn from AI, facilitated by XAI, helps enhance their clinical judgment. Human experts actively engage in validating AI predictions through AI interrogation practices, investing time and analytical effort to align their claims with AI insights. This is particularly vital when experts' judgments deviate from AI predictions, as XAI plays a key role in bridging knowledge gaps. It enables experts to reconcile their understanding with that of the AI and expand their knowledge base through these interrogation practices. In doing so, our study engages with and contributes to the organizational literature on effective collaboration in knowledge work (Carlile, 2004; Levina, 2005) by extending the discussion to human-AI collaboration.

This research contributes to the ongoing discussion on opacity (Griffin and Grote, 2020; Oberkampf et al., 2004; Packard and Clark, 2020) by showing that AI opacity increases human experts' epistemic uncertainty, leading them to enact AI interrogation practices. We advance this discourse by incorporating XAI, demonstrating how it can reduce some of the uncertainties associated with opaque AI systems. In doing so, we also contribute to discussions on XAI and epistemic uncertainty (Jiang et al., 2022; Tomsett et al., 2020). Previous research shows that AI opacity can hamper user acceptance of the technology and trigger resistance and algorithm aversion, especially when AI outputs contradict human experience and intuition (Bauer et al., 2021; Ribeiro et al., 2016). This makes the provision of explanations crucial to preventing tension between humans and machines. The introduction of XAI enhances trust in the AI system and promotes a willingness to learn from it. .

The current debate in the literature on AI explainability (e.g., Bauer et al., 2023; Erlei et al., 2020) and AI opacity (e.g., Waardenburg et al., 2022) suggests two primary approaches: limiting AI use in critical decisions when transparency is unattainable, and designing XAI systems to enhance transparency. Lebovitz et al. (2022) proposed a third approach, in which professionals use AI interrogation practices to manage AI opacity and validate its results. We contribute to this discourse by examining how experts deal with AI opacity and extend the scope to explore how they deal with AI explainability. We find that experts engage in interrogative practices to address opacity and continue these practices— along with other responses— even after the introduction of XAI, in order to validate AI outcomes. These findings suggest that experts will continue to enact interrogative practices regardless of whether the AI system is explainable or functions as a black box, indicating that humans take an active rather than passive role when collaborating with AI rather than a passive one.

Furthermore, the introduction of XAI has led clinicians to recognize the potential of AI systems to identify nuances that might elude human detection, remember infants' movements more reliably than humans, and offer standardized evaluations. These standardized evaluations can mitigate different assessments that arise from the subjective nature of CP assessment, thus reducing variability and ensuring that different practitioners evaluating the same patient are less likely to reach inconsistent conclusions. However, such standardization may also foster learning myopia—such as neglecting long-term outcomes, overlooking failures, and reinforcing inflexibility and other biases (Levinthal and March, 1993). This occurs because AI-generated standardized results reduce diversity in routines anddiminish the richness of contextual and causal knowledge within organizations, both of which are essential for organizational learning (Balasubramanian et al., 2022). In the context of this study, we argue that reliance on AI for standardized evaluations can lead to learning myopia within organizations by reducing diversity in routines and depth of knowledge. These findings contribute to the extant scientific discussion on learning in the age of AI, where overreliance on AI-driven recommendations is increasingly recognized as a potential hindrance to both individual and organizational learning. Such overreliance may reduce interactions among individuals, which are crucial for both individual-level learning—shaped by personal experiences and responses to changes—and for organizational learning, fostered through interactions among individuals, artifacts, and the environment (Balasubramanian et al., 2022). Consequently, the standardized evaluations provided by AI may limit both individual and organizational learning capacity..

A similar pattern of disrupted learning has been observed with the integration of robotic systems in surgical settings, where reduced direct human interaction has profoundly impacted traditional learning structures. For instance, Beane (2019) observed that the lack of direct interaction between trainees and experienced surgeons has compelled trainees to adopt self-directed—and often less effective—learning strategies, yielding mixed results. Similarly, robotic systems in surgery have disrupted traditional learning by displacing residents from active participation, relegating them to observational or minor roles, thereby eroding their ability to acquire critical surgical skills (Sergeeva et al., 2020). This shift from hands-on tasks to passive observation in the robotic setups has not only limited learning opportunities and but also weakened the collaborative team interactions essential for effective surgical training.

This broader pattern of technology influencing human interaction and knowledge acquisition is not limited to surgical settings. Current research suggests that AI advice causes individuals to converge towards similar responses, increasing overall accuracy but reducing unique human knowledge and diversity (Fügener et al., 2021). This behavioral convergence negatively impacts collaborative decision-making, as people tend to mirror AI recommendations rather than engage in independent, critical thinking or creative problem-solving. Similarly, excessive reliance on Large Language Models (LLMs) raises concerns about epistemological diversity and the role of human involvement in knowledge production (Cornelissen et al., 2024). LLMs, driven by existing data, inherently prioritize established patterns and perspectives (e.g., a single form of epistemology), often marginalizing alternatives and limiting diversity.

Furthermore, overreliance on AI systems exacerbates this challenge by limiting exposure to diverse perspectives. While AI systems enhances coordination, they typically focus on specific tasks or domains by curating and prioritizing information. This targeted approach confines individuals to narrower ways of thinking, as they primarily engage with knowledge within a limited scope, reducing their ability to access diverse viewpoints (Faraj et al., 2018). Without such diversity, opportunities for knowledge recombination and the emergence of serendipitous or innovative solutions are significantly diminished. Thus, while AI systems offer short-term gains in efficiency and productivity, these benefits must be weighed against their long-term effects on learning, knowledge sharing, and professional expertise.

Finally, our findings show that explicitly articulating AI's certainty levels in its predictions is essential for fostering trust among experts. This aligns with current research suggesting that presenting AI certainty improves human understanding of the AI's error boundaries (Bansal et al., 2019). Providing this infromation can help reduce overreliance on AI, as individuals are better equipped to challenge incorrect AI predictions. Fügener et al. (2021) found that presenting AI certainty can actually reduce blind trust in AI, making individuals less likely to follow AI adviceunquestioningly and more capable of disregarding incorrect suggestions—thus mitigating the reduction in unique human knowledge.

5.3. Practical implications

Our findings reveal that clinicians may doubt their own competence when their judgments contradict AI predictions. Depending on the clinical context, the disease in question, and the potential consequences of incorrect medical treatment, such doubt can have significant implications—particularly in cases where AI makes false positive or false negative predictions. In our study, where the AI assessment tool is one of several diagnostic aids, a false positive or negative result carries limited consequences. However, in other scenarios,such as cancer diagnosis,AI-based decision-making could lead to more severe consequences. Therefore, the interaction between human experts and AI must be carefully managed to ensure that final decisions always remain with humans. While AI can support or even automate decisions (Gillner, 2024) in the context of AI-assisted diagnoses, the final responsibility typically rests with healthcare professionals, who rely on their own professional experience and clinical assessment tools (Lupton and Jutel, 2015).

Our findings reveal a desire among clinicians to learn from AI, suggesting that management should design an environment that encourages effective knowledge sharing between humans and AI while avoiding excessive reliance on the technology. The current AI system under consideration lacks the ability to adjust to feedback and preferences from the users, resulting in a unidirectional interaction. In this context, the AI system can challenge human claims by highlighting discrepancies and presenting alternative claims, encouraging clinicians to critically assess their judgments. Additionally, management could develop programs, particularly for less experienced clinicians to support learning from AI's predictions through the lens of explainability. Such programs should emphasize the importance of critically evaluating AI, questioning its assumptions, and understanding the potential for false positive or false negative predictions.

XAI enables achieving transparency, which is essential for enhancing trust and accountability in critical settings such as healthcare. When these systems provide clear insights into their decision-making and predictions processes, medical professionals can better understand and validate their recommendations. This understanding not only gives the decision makers insight into how the system works, but also fosters greater engagement with the AI system, by enablingmore interaction and critical evaluation of its outputs, rather than viewing it as a categorical tool. Such, transparency can help prevent misdiagnoses and ensure that clinical decisions are informed and reliable. By enabling a deeper understanding of the underlying logic, explainable technologies create an environment where medical practitioners and AI systems can work together to enhance patient care. This collaboration ultimately benefits society by making the use of AI in healthcare more trustworthy.

To this end, we argue that XAI explanations should enhance technical transparency and enable actionable, and meaningful collaboration between end-users and AI systems. By helping experts engage with and reflect on AI predictions, XAI can support more informed judgment formation and encourage thoughtful, and interactive engagement. Our findings suggest that the design of XAI features, and human-AI interactions must evolve to better address end-user needs.

5.3.1. Guidelines for designing human-AI interaction

Drawing on our empirical findings, we propose a set of guidelines for designing human-AI interactions to support judgment formation. Informed by insights from our data, these guidelines aim to maximize the advantages of human-AI collaboration.

5.3.1.1. Promoting user-centered AI development. Our study highlights the importance of designing AI systems that complement rather than replace, human expertise. Therefore, involving users and encouraging them to provide input and feedback is crucial for developing software that aligns with clinical workflows and decision-making processes is crucial. For example, in this study, the users provided feedback on their understanding of how the colors on body parts were interpreted. Such feedback can offer valuable guidance for developing software that is intuitively usable by experts. Bidirectional human-centric interaction—where users contribute their feedback and preferences— supports iterative improvements and enables greater acceptance of the AI system.

5.3.1.2. Developing meaningful explainability. Insight into the AI system's decision-making process— such as the localization of the important infant body areas for the CP prediction, in our case— can increase user trust *if* the explanation aligns with the user's understanding and enables the user to understand the driver behind the system's prediction. Whether this occurs depends not only on the XAI method itself but also on details of the presentation—in our case the illustration, of the generated explanation. Thus, the requirement for the XAI method is twofold: the explanation must be faithful to the model, see also (Pellano et al., 2024b) and it must be represented in a way that is intuitive to the end-user. These two aspects are, in principle,independent, but both must be optimized in order for the explanation to be meaningful overall.

5.3.1.3. Enabling discovery and learning in interaction. Designing systems with features that trigger exploration can enhance users' ability to engage with and understand the system functionalities. In this study, we implemented a design that continuously displayed risk values and 5-s segments of high- and low-risk moments, synchronized with video footage of the infant. This setup sparked curiosity and encouraged users to explore the relationship between specific movements and the corresponding risk values. Aligned with (Miller, 2023), we propose that systems should be designed to encourage users to engage in critical reflection, prompting them to consider whether they may need to revise their own judgment, identify potential oversights in their reasoning, or recognize possible errors in the model's output. Explanations generated by XAI should not solely function as a justification for the model's predictions or as a tool to persuade the user.

5.3.1.4. Presenting AI certainty levels for building trust. Our data reveals that a crucial factor in establishing trust in AI predictions is the clear communication of certainty levels for each case. AI systems should be designed to transparently convey the degree of confidence in their predictions— for instance, by providing certainty scores derived from consensus across multiple models, as we demonstrated in our experimental setting using 70 different models. In our study, clinicians were presented with predictions accompanied by certainty levels and accuracy data, which influenced their willingness to rely on the AI prediction to form their judgment. We argue that, by integrating certainty levels into the system design, AI can better support experts in making more informed decisions and build trust in the system's reliability. This, in turn, helps clinicians gauge how much to rely on the AI without undermining their own confidence, especially when the AI indicates low certainty in its predictions.

5.3.1.5. Promoting independent judgment formation and preserving human interaction. Our findings highlight the importance of preventing learning myopia, a risk associated with over-reliance on AI predictions.

To mitigate this, we propose that human-AI interaction should be designed to prioritize the development of independent human judgment. For instance, experts in our study preferred forming their initial judgment before being exposed to AI predictions, to avoid being influenced by AI outputs. In addition, designing environments where feedback from human colleagues is readily available enhances the diversity of perspectives. These interactions with peers introduce diverse viewpoints and challenge standardized AI conclusions. Discussions with colleagues provide experts with access to a broader range of insights and help them avoid depending solely on AI predictions. We argue that integrating these practices into human-AI interaction design supports more informed judgment formation. In doing so, AI is positioned as a complementary tool, rather than one that diminishes professional expertise through over-reliance.

5.3.1.6. Presenting training data diversity and performance metrics. Our study revealed that providing sensitivity and specificity metrics about the AI system's accuracy to human experts plays a pivotal role in building trust in AI systems. Notably, some experts valued these metrics even more than the explainability of the AI, highlighting them as key factors in establishing trust. Additionally, transparency about the training data— such as dataset size and geographic origin— further enhances confidence in the model's representativeness and reliability. This transparency allows the model to be applied across diverse scenarios.

5.4. Limitations and future research

While our study contributes to the literature on XAI and human-AI collaboration in healthcare, it has limitationsthat warrant further investigation. First, we used scenario-based interviews; future studies could conduct experiments in realclinical settings where clinicians make judgments based on actual cases alongside AI predictions. This would allow reseachers to observe clinicians responses in situations with immediate consequences. This approach is important because previous research indicates that humans may overrule AI claims (Lebovitz et al., 2022), which we did not observe this— possibly due to the absence of real-life consequences in our scenarios. Future research should examine whether this absence is attributable to the experimental setup, perceptions of machine intelligence, or over-trust in AI. Investigating the perception of machine intelligence and over-trust in AI is crucial, as higher perceived intelligence levels can lead to greater trust, even when the system's reliability is low (Glikson and Woolley, 2020). Second, our study may have included a limited number of clinicians from specific demographics. Future research should expand the sample size and include a more diverse group to enhance the generalizability of the findings.

Future research could also compare decision-making and outcomes in clinical settings where clinicians use AI assistance versus relying solely on their expertise. This approach would help in clarify the real-world impact of AI on clinical decision-making. Such research is important given the common assumption that AI augmentation can help organizations achieve positive outcomes by improving knowledge, insights, and efficiency (Ogunbiyi et al., 2021; Raisch and Krakowski, 2021; Wilson and Daugherty, 2018). AI systems, especially XAI, may reduce certain human errors by identifying their sources and minimizing their occurrence (Kohl et al., 2019). However, they can also introduce new types of errors due to biases or poor-quality training data (Lebovitz et al., 2022). Since the quality of AI outputs depends heavily on the training data, these limitations are particularly concerning in healthcare decision-making, where they may lead to misleading claims.

Our findings reveal that upon receiving XAI insights, experts began to establish connections with the AI by following different paths. While we have explored these processes theoretically in the findings and discussion sections, our dataset does not capture the specific rationales and factors underlying each path. We recognize this as a limitation of our study and suggest it as an important avenue for future research to enhance and extend our understanding. Furthermore, given that the AI system in our study enabled unidirectional feedback—prompting users to reassess their judgments based on AI predictions— future research could explore or design AI systems that support bidirectional interaction. Such a design would allow both parties to challenge each other's claims, which would create an environment of mutual learning between experts and AI. Additionally, future research could examine the psychological effects of using AI on experts, particularly how it impacts their confidence and perceived competence. This could help inform the design of interventions that support experts in maintaining their expertise while benefiting from AI assistance.

Furthermore, future research is warranted to examine how individual factors shape trust in AI systems beyond system-related properties such as opacity and explainability. Prior research has identified several relevant professional characteristics— including self-confidence, preferences for control, personal expectations, risk tolerance, cultural background, and domain expertise— as important factors influencing trust (Glikson and Woolley, 2020). For instance, professionals with higher self-confidence or greater expertise tend to rely less on AI advice, even when doing so may improve decision outcomes (Glikson and Woolley, 2020; Logg et al., 2019). Moreover, our empirical findings suggest that low AI literacy—defined as the user's understanding of basic AI functions and how AI systems operate, can lead to overtrust. For example, several participants expressed greater confidence in the system simply because they learned it was trained on over 400 videos. This information appeared to be treated as equivalent to human experience with 400 clinical cases. Such perception reflects a superficial understanding of how AI learns. Although the size of the training data matters, actual AI performance also depends on data quality, diversity, relevance, and other technical factors that are often not visible to users. Thus, further research should examine how different levels of AI literacy influence trust. Such research could inform targeted training initiatives as well as system design practices that help professionals calibrate their trust in AI systems.

Finally, though this study primarily addresses epistemic uncertainty—arising from limited knowledge and the opacity of AI predictions—it also opens the door to future research on qualitative uncertainty. This form of uncertainty stems from subjectivity and includes factors such as the context of data collection, the assumptions inherent in methods, and the conceptual or epistemological lenses through which knowledge is interpreted (Panagiotidou and Vande Moere, 2022; Kennedy, 2013). As Chen et al. (2007) explain, qualitative uncertainty arises from "subjective and biased beliefs" and encompasses conceptual, methodological, epistemological, symbolic, and linguistic elements. Future research could explore how human experts handle situations where their own interpretations or experiences conflict with the reasoning behind AI outputs. These disagreements may not stem from missing information but rather from differences between the expert's perspective or understanding and the underlying logic embedded in the AI— such as how data was selected, categorized, or prioritized during system development.

6. Conclusion

In this study, we explored how experts respond to AI opacity and how XAI reshapes judgment formation in high-stakes clinical contexts. By examining experts' interactions with opaque AI predictions, we found that opacity increased epistemic uncertainty, prompting experts' either to critically revisit their initial judgments or to engage in deliberate AI interrogation practices. In particular, epistemic uncertainty drove experts towards rigorous interrogation practices aimed at resolving contradictions between algorithmic outputs and their professional judgments.

Our analysis suggests that the introduction of XAI transforms this uncertainty into a productive dialogue, leading to deeper engagement, increased trust, and greater openness to AI contributions. Consequently, the experts were more inclined to reconsider or update their initial judgments when discrepancies arose between their assessments and the AI's predictions. Notably, our findings also indicated that the experts maintained a cautious stance, explicitly cautioning against exclusive reliance on AI tools for clinical decision-making.

This research contributes to the organizational and information systems literatures by illuminating how AI opacity affects human-AI interactions. It highlights the beneficial role of explainability in enhancing experts' judgment formation and trust in AI systems. Nevertheless, we also emphasize experts' caution against exclusive dependence on AI systems. Although explainability clearly alleviates uncertainty and supports enhanced human-AI collaboration, it remains crucial to ensure that AI systems are not over-relied upon in general, as this could potentially hinder experts' ongoing critical reflection and professional judgment.

CRediT authorship contribution statement

Reda Hassan: Writing – review & editing, Writing – original draft, Resources, Investigation, Conceptualization, Visualization, Project administration, Formal analysis, Validation, Methodology, Data curation. **Nhien Nguyen:** Visualization, Formal analysis, Writing – original draft, Writing – review & editing, Validation, Data curation, Methodology, Conceptualization. **Stine Rasdal Finserås:** Investigation, Conceptualization, Project administration, Data curation, Writing – review & editing, Formal analysis. **Lars Adde:** Writing – review & editing, Conceptualization, Funding acquisition, Resources. **Inga Strümke:** Conceptualization, Writing – review & editing. **Ragnhild Støen:** Writing – review & editing, Conceptualization, Funding acquisition, Resources.

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Data availability

The data that has been used is confidential.

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